Luca **Corbucci**

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About Me

Postdoctoral Researcher in Multimodal LLMs at FBK (Trento, Italy). I hold a PhD in Computer Science focused on AI Safety for Federated Learning. Authored 5+ publications in top-tier venues, including **FAccT** and **ECAI**. Proven leader and collaborator: successfully initiated a 1-year research internship with **Telefónica Research Lab** and projects with **Johannes Gutenberg-Universität Mainz**. Experienced public speaker, community builder (FOSDEM AI Devroom, Superhero Valley, Pisa.dev), and **podcaster** (280+ episodes).

Work Experience

Fondazione Bruno Kessler - Machine Translation Unit

Trento, Italy

Postdoctoral Researcher - Supervised by Dr. L. Bentivogli — *Mar. 2026 - Ongoing*

- Worked on the training and evaluation of SpeechLLMs using the Nemo Framework.
- Contributed to European research projects, including DVPS, Meetween, and LLMs4EU.

Telefónica R&D - System AI Lab

Barcelona, Spain

Research Intern - Supported by the SPATIAL H2020 EU project and supervised by Dr. Nicolas Kourtellis — *Mar. 2023 - Mar. 2024*

- Designed PUFFLE, a privacy-preserving **Federated Learning** method achieving 75% fairness improvement with minimal (22%) accuracy loss.
- Integrated fairness constraints based on practitioner requirements, bridging the gap between theoretical fairness and real-world systems.
- **Paper** accepted at the European Conference on Artificial Intelligence (**ECAI**) 2024; **Patent** application submitted.

Education

University of Pisa

Pisa, Italy

PhD in Computer Science - **Advisors:** Prof. A. Monreale and Dr. N. Kourtellis. - **Final Grade:** Summa Cum Laude — *Nov. 2021 - Jul. 2025*

- Developed methodologies to ensure privacy and fairness in Federated Learning models, addressing key challenges in **safe and ethical AI**.
- Explored and designed techniques for explainable Federated Learning (**XAI** for FL) to enhance model transparency and trust.
- Started collaboration with **Telefonica Research Lab** (1-year visit) and **Johannes Gutenberg-Universität Mainz**.
- Partially funded and supported by the Horizon Europe project "TANGO".

University of Pisa

Pisa, Italy

M.Sc. in Computer Science - **Final Grade:** 110/110 Cum Laude — *Oct. 2018 - Jun 2021*

- **Thesis:** "Semantic enrichment of XAI explanations for healthcare". [**Paper** | **Code**].
- Spent a semester at **Technical University of Munich** thanks to the Erasmus+ study program.

University of Pisa

Pisa, Italy

B.Sc. in Computer Science — *Sep. 2014 - Oct. 2018*

- **Thesis:** "Construction and Analysis of the Ethereum Transaction Graphs". [**Code**].

Selected Publications - Full list: Google Scholar

Sep 2026	**L. Corbucci**, Marco Gaido, Sara Papi, Yiu Chung Leung, Matteo Negri, Luisa Bentivogli, *"ASR Data is All You Need to Build Instruction-Following SpeechLLMs" - Under Review*	
Sep 2026	A. Ercolani, E. Federici, L. Bentivogli, **L. Corbucci**, *"Training Small LLMs from Scratch: Lessons from Zagreus and Nesso" - Accepted at CLiC-it 2026*	*Palermo, Italy*
Jun 2026	X. Heilmann, **L. Corbucci**, M. Cerrato, *"Rashomon Sets and Model Multiplicity in Federated Learning" - Accepted at ACM FAccT 2026*	*Montréal, Canada*
Jun 2026	X. Heilmann, **L. Corbucci**, M. Cerrato and A. Monreale, *"FeDa4Fair: Client-Level Federated Datasets for Fairness Evaluation." - Accepted at ACM FAccT 2026*	*Montréal, Canada*
Jun 2025	**L. Corbucci**, X. Heilmann, M. Cerrato, *"Benefits of the Federation? Analyzing the Impact of Fair Federated Learning at the Client Level" - Accepted at ACM FAccT 2025*	*Athens, Greece*

Community & Leadership

Superhero Valley

Pisa, Italy

Co-founder and Community Manager — *Oct. 2020 - Ongoing*

- Co-founded a community of **2000+** computer scientists supported by big-tech employees and Italian universities
- Supervised a team of **5** helpers in organising community activities and events.
- **80+** community members got internships or full-time positions in Big Tech companies thanks to our activities.

PointerPodcast

Pisa, Italy

Co-Creator and Podcaster — *May. 2019 - Ongoing*

- Co-created a podcast with **250+** episodes, **70+** guests from industry and academia, and **40k+** total listens.